

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=7,count=0;
6
7         for(int i=1;i<=n;i++)
8             if(n%i==0)
9                 count++;
10
11         if(count==2)
12             System.out.println("Prime");
13         else
14             System.out.println("Not Prime");
15     }
16 }
```

prime

..Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=8,count=0;
6
7         for(int i=1;i<=n;i++)
8             if(n%i==0)
9                 count++;
10
11         if(count>2)
12             System.out.println("Composite");
13         else
14             System.out.println("Not Composite");
15     }
16 }
```

Composite

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int year=2024;
6
7         if(year%4==0)
8             System.out.println("Leap Year");
9         else
10            System.out.println("Not Leap Year");
11     }
12 }
13
```



Leap Year

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=153,temp=n,sum=0,r;
6
7         while(n>0)
8         {
9             r=n%10;
10            sum=sum+r*r*r;
11            n=n/10;
12        }
13
14        if(sum==temp)
15            System.out.println("Armstrong");
16        else
17            System.out.println("Not Armstrong");
18    }
19 }
```

Armstrong

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a=0,b=1,c;
6
7         System.out.print(a+" "+b+" ");
8
9         for(int i=1;i<=8;i++)
10        {
11            c=a+b;
12            System.out.print(c+" ");
13            a=b;
14            b=c;
15        }
16    }
17 }
```

0 1 1 2 3 5 8 13 21 34

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=1234,r,rev=0;
6
7         while(n>0)
8         {
9             r=n%10;
10            rev=rev*10+r;
11            n=n/10;
12        }
13
14        System.out.println(rev);
15    }
16 }
```

4321

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=12;
6
7         for(int i=1;i<=n;i++)
8             if(n%i==0)
9                 System.out.print(i+" ");
10    }
11 }
```



```
1 2 3 4 6 12
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=10,sum=0;
6
7         for(int i=1;i<=n;i++)
8             sum=sum+i;
9
10        System.out.println(sum);
11    }
12 }
```



55

```
...Program finished with exit code 0
Press ENTER to exit console.
```



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a[]={1,2,3,2,4};
6
7         for(int i=0;i<a.length;i++)
8         {
9             for(int j=i+1;j<a.length;j++)
10            {
11                if(a[i]==a[j])
12                    System.out.println(a[i]);
13            }
14        }
15    }
16 }
```

input

```
2
...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=10;i+=2)
6             System.out.print(i+" ");
7     }
8 }
```



1 3 5 7 9

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a=12,b=18,gcd=1;
6
7         for(int i=1;i<=a&& i<=b;i++)
8             if(a%i==0 && b%i==0)
9                 gcd=i;
10
11         System.out.println(gcd);
12     }
13 }
```



```
5
...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a=12,b=18,gcd=1,lcm;
6
7         for(int i=1;i<=a&& i<=b;i++)
8             if(a%i==0 && b%i==0)
9                 gcd=i;
10
11         lcm=(a*b)/gcd;
12
13         System.out.println(lcm);
14     }
15 }
```

input

36

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a=0,b=1,c=0,n=8;
6
7         for(int i=2;i<=n;i++)
8         {
9             c=a+b;
10            a=b;
11            b=c;
12        }
13
14        System.out.println(c);
15    }
16 }
```

input

21

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a[]={1,2,2,3,2};
6         int count=0;
7
8         for(int i=0;i<a.length;i++)
9             if(a[i]==2)
10                count++;
11
12         System.out.println(count);
13     }
14 }
```

input

3
...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=5;i++)
6         {
7             for(int j=1;j<=i;j++)
8                 System.out.print("*");
9
10            System.out.println();
11        }
12    }
13 }
```

input

```
*
**
***
****
*****

...Program finished with exit code 0
Press ENTER to exit console.
```

```

1  class Main
2  {
3      public static void main(String[] args)
4      {
5          for(int i=1;i<=5;i++)
6          {
7              for(int j=1;j<=5-i;j++)
8                  System.out.print(" ");
9
10             for(int k=1;k<=2*i-1;k++)
11                 System.out.print("*");
12
13             System.out.println();
14         }
15     }
16 }

```

input

```

*
***
*****
*****
*****
.....Program finished with exit code 0
Press ENTER to exit console.

```



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=5;i++)
6         {
7             for(int j=1;j<=i;j++)
8                 System.out.print("*");
9
10            System.out.println();
11        }
12
13        for(int i=4;i>=1;i--)
14        {
15            for(int j=1;j<=i;j++)
16                System.out.print("*");
17
18            System.out.println();
19        }
20    }
21 }
```

input

```
*
**
***
****
*****
*****
****
***
**
*

...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=5;i>=1;i--)
6         {
7             for(int j=1;j<=5-i;j++)
8                 System.out.print(" ");
9
10            for(int k=1;k<=2*i-1;k++)
11                System.out.print("*");
12
13            System.out.println();
14        }
15    }
16 }
```

input

```
*****
*****
*****
***
*
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=5;i++)
6         {
7             for(int j=1;j<=5;j++)
8             {
9                 if(i==1||i==5||j==1||j==5)
10                    System.out.print("*");
11                 else
12                    System.out.print(" ");
13             }
14             System.out.println();
15         }
16     }
17 }
```

input

```
*****
*   *
*   *
*   *
*****
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=5;i>=1;i--)
6         {
7             for(int j=1;j<=i;j++)
8                 System.out.print("*");
9
10            System.out.println();
11        }
12    }
13 }
```

input

```
*****
*****
****
***
**
*

...Program finished with exit code 0
Press ENTER to exit console.
```

```

1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=5;i++)
6         {
7             for(int j=1;j<=5-i;j++)
8                 System.out.print(" ");
9
10            for(int k=1;k<=2*i-1;k++)
11                System.out.print("*");
12
13            System.out.println();
14        }
15
16        for(int i=4;i>=1;i--)
17        {
18            for(int j=1;j<=5-i;j++)
19                System.out.print(" ");
20
21            for(int k=1;k<=2*i-1;k++)
22                System.out.print("*");
23
24            System.out.println();
25        }
26    }
27 }

```

input

```

*
***
*****
*****
*****
*****
*****
***
*

```

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=5;i++)
6         {
7             for(int j=1;j<=5-i;j++)
8                 System.out.print(" ");
9
10            for(int k=1;k<=5;k++)
11                System.out.print("*");
12
13            System.out.println();
14        }
15    }
16 }
```

input

```
*****
 *****
  *****
   *****
    *****
 ..Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=5;i++)
6         {
7             for(int j=1;j<=5;j++)
8                 System.out.print("*");
9
10            System.out.println();
11        }
12    }
13 }
```

input

```
*****
*****
*****
*****
*****

...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a[][]={{1,2},{3,4}};
6         int b[][]={{5,6},{7,8}};
7         int c[][]=new int[2][2];
8
9         for(int i=0;i<2;i++)
10        {
11            for(int j=0;j<2;j++)
12            {
13                c[i][j]=a[i][j]+b[i][j];
14                System.out.print(c[i][j]+" ");
15            }
16            System.out.println();
17        }
18    }
19 }
```

input

```
6 8
10 12
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         String s="TEMPLE";
6
7         for(int i=s.length()-1;i>=0;i--)
8             System.out.print(s.charAt(i));
9     }
10 }
```

ELPMET

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         String s="1234";
6
7         int n=Integer.parseInt(s);
8
9         System.out.println(n);
10    }
11 }
```

input

1234

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         String s1="Saveetha@789";
6         String s2="Saveetha@123";
7
8         if(s1.equals(s2))
9             System.out.println("User name is Valid");
10        else
11            System.out.println("User name is Invalid");
12    }
13 }
```

input

User name is Invalid

...Program finished with exit code 0
Press ENTER to exit console.

```
1 import java.util.Arrays;
2
3 class Main
4 {
5     public static void main(String[] args)
6     {
7         String a[]={"Banana","Carrot","Radish","Apple","Jack"};
8
9         Arrays.sort(a);
10
11         for(String i:a)
12             System.out.println(i);
13     }
14 }
```

Apple
Banana
Carrot
Jack
Radish

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         String s="Java@123#";
6         int count=0;
7
8         for(int i=0;i<s.length();i++)
9         {
10             char ch=s.charAt(i);
11
12             if(!Character.isLetterOrDigit(ch))
13             {
14                 System.out.print(ch+" ");
15                 count++;
16             }
17         }
18
19         System.out.println("\nCount = "+count);
20     }
21 }
```

Input

```
@ #
Count = 2

...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         String s="Saveetha School of Engineering";
6         int count=0;
7
8         s=s.toLowerCase();
9
10        for(int i=0;i<s.length();i++)
11        {
12            char ch=s.charAt(i);
13
14            if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
15                count++;
16        }
17
18        System.out.println(count);
19    }
20 }
```

input

```
12
...Program finished with exit code 0
Press ENTER to exit console.
```

```

1 class Main
2 {
3     public static void main(String[] args)
4     {
5         String s="Engineering";
6
7         s=s.toLowerCase();
8
9         System.out.print("Vowels: ");
10
11        for(int i=0;i<s.length();i++)
12        {
13            char ch=s.charAt(i);
14
15            if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
16                System.out.print(ch+" ");
17        }
18
19        System.out.print("\nConsonants: ");
20
21        for(int i=0;i<s.length();i++)
22        {
23            char ch=s.charAt(i);
24
25            if(Character.isLetter(ch) && !(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u'))
26                System.out.print(ch+" ");
27        }
28    }
29 }

```

input

```

Vowels: e i e e i
Consonants: n g n r n g

...Program finished with exit code 0
Press ENTER to exit console.

```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         double n=0.6;
6
7         System.out.println("Square = "+(n*n));
8         System.out.println("Cube = "+(n*n*n));
9     }
10 }
```



```
Square = 0.36
Cube = 0.216
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=7;
6
7         for(int i=1;i<=n;i=i+2)
8             System.out.print(i+" ");
9     }
10 }
```



1 3 5 7

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=143,sum=0;
6
7         while(n>0)
8         {
9             sum=sum+n%10;
10            n=n/10;
11        }
12
13        while(sum>9)
14        {
15            int t=0;
16
17            while(sum>0)
18            {
19                t=t+sum%10;
20                sum=sum/10;
21            }
22
23            sum=t;
24        }
25
26        System.out.println(sum);
27    }
28 }
```

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=6561;
6
7         int r=(int)Math.sqrt(n);
8
9         System.out.println(r);
10        System.out.println(-r);
11    }
12 }
```



```
31
-81
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=27;
6
7         while(n>1 && n%3==0)
8             n=n/3;
9
10        if(n==1)
11            System.out.println("True");
12        else
13            System.out.println("False");
14    }
15 }
```

True

...Program finished with exit code 0
Press ENTER to exit console.

```

1 import java.util.*;
2
3 class Main
4 {
5     public static void main(String[] args)
6     {
7         String s="Ram hit a ball the hit ball flew far after it was hit";
8
9         String ban="hit";
10
11         String a[]=s.toLowerCase().split(" ");
12
13         HashMap<String,Integer> map=new HashMap<>();
14
15         for(String i:a)
16         {
17             if(!i.equals(ban))
18                 map.put(i,map.getOrDefault(i,0)+1);
19         }
20
21         String ans="";
22         int max=0;
23
24         for(String i:map.keySet())
25         {
26             if(map.get(i)>max)
27             {
28                 max=map.get(i);
29                 ans=i;
30             }
31         }
32
33         System.out.println(ans);
34     }
35 }

```

input

ball

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a[]={1,0,2,3,0,4,5,0};
6
7         for(int i=0;i<a.length;i++)
8         {
9             System.out.print(a[i]+" ");
10
11             if(a[i]==0)
12                 System.out.print("0 ");
13         }
14     }
15 }
```

1 0 0 2 3 0 0 4 5 0 0

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a[]={3,0,1};
6
7         int n=a.length;
8
9         int sum=n*(n+1)/2;
10
11         int s=0;
12
13         for(int i=0;i<n;i++)
14             s=s+a[i];
15
16         System.out.println(sum-s);
17     }
18 }
```

input

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=5;
6
7         for(int i=0;i<n;i++)
8         {
9             int num=1;
10
11             for(int j=0;j<=i;j++)
12             {
13                 System.out.print(num+" ");
14                 num=num*(i-j)/(j+1);
15             }
16             System.out.println();
17         }
18     }
19 }
```



```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=4;i++)
6         {
7             for(int j=1;j<=6;j++)
8                 System.out.print("# ");
9
10            System.out.println();
11        }
12    }
13 }
```



```
# # # # #
# # # # #
# # # # #
# # # # #
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int n=3;
6
7         for(int i=1;i<=n;i++)
8         {
9             for(int j=1;j<=i;j++)
10                System.out.print("1");
11
12                System.out.println();
13            }
14
15            for(int i=n-1;i>=1;i--)
16            {
17                for(int j=1;j<=i;j++)
18                    System.out.print("1");
19
20                System.out.println();
21            }
22        }
23    }
```

1
11
11
11
1

...Program finished with exit code 0
Press ENTER to exit console.

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int m=4,n=5;
6
7         for(int i=1;i<=n;i++)
8             System.out.println(i+" x "+m+" = "+(i*m));
9     }
10 }
```

input

```
1 x 4 = 4
2 x 4 = 8
3 x 4 = 12
4 x 4 = 16
5 x 4 = 20

...Program finished with exit code 0
Press ENTER to exit console.
```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=5;i++)
6         {
7             for(int j=1;j<=5;j++)
8             {
9                 if(i==1||i==5||j==1||j==5)
10                    System.out.print("$ ");
11                 else
12                    System.out.print(" ");
13             }
14            System.out.println();
15        }
16    }
17 }
```

input

```
$ $ $ $ $
$      $
$      $
$      $
$ $ $ $ $

...Program finished with exit code 0
Press ENTER to exit console.
```

```

1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=4;i++)
6         {
7             for(int j=1;j<=i;j++)
8                 System.out.print(i+" ");
9
10            System.out.println();
11        }
12
13        for(int i=3;i>=1;i--)
14        {
15            for(int j=1;j<=i;j++)
16                System.out.print(i+" ");
17
18            System.out.println();
19        }
20    }
21 }

```

input

```

1
2 2
3 3 3
4 4 4 4
3 3 3
2 2
1

...Program finished with exit code 0
Press ENTER to exit console.

```

```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         for(int i=1;i<=4;i++)
6         {
7             for(int j=1;j<=i;j++)
8                 System.out.print(i+" ");
9
10            System.out.println();
11        }
12    }
13 }
```

input

```
1
2 2
3 3 3
4 4 4 4

...Program finished with exit code 0
Press ENTER to exit console.
```